

Abhay Singh Sengar

Bhopal, Madhya Pradesh, India

+91 7898551977 — sengarabhay03@gmail.com — LinkedIn — GitHub

Professional Summary

Computer Science undergraduate focused on Machine Learning and Generative AI systems, experienced in building multi-agent architectures and retrieval-based pipelines. Strong interest in model evaluation, reliability, and scalable AI deployment. Seeking AI/ML Internship or Entry-Level AI Engineer roles.

Experience

Google Cloud GenAI Virtual Internship — Smart Bridge

Nov 2025 – Jan 2026 Remote

- Developed proof-of-concept Retrieval-Augmented Generation (RAG) pipelines on Google Cloud Vertex AI to ground LLM outputs in domain-specific datasets.
- Integrated Gemini APIs into structured workflows emphasizing retrieval-first design for improved factual reliability.
- Engineered query pipelines that reduced hallucination risk compared to naive prompting strategies.
- Worked with vector search infrastructure and production-style cloud AI architecture patterns.

Selected Projects

Multi-Agent DSA Solver *Python • Multi-Agent Systems • LLM Orchestration*

- Architected a multi-agent system capable of decomposing complex data structures and algorithms problems into structured reasoning stages.
- Designed orchestration pipelines coordinating agents responsible for interpretation, planning, code synthesis, and validation.
- Implemented modular communication protocols enabling dynamic task delegation and improved reasoning coherence.
- Enhanced logical consistency through enforced step-wise reasoning, outperforming single-pass LLM approaches.

Multimodal RAG System for Intelligent Document Understanding *Python • LangChain • FAISS • OCR*

- Built an end-to-end multimodal retrieval system capable of answering queries across documents containing text and visual elements.
- Engineered separate embedding pipelines for textual and visual modalities to significantly improve retrieval precision.
- Designed modality-aware query routing to prevent irrelevant context injection and reduce hallucinated responses.
- Generated structured outputs supporting downstream analytics and automated visualization workflows.

Xtractor — Multi-Agent Knowledge Extraction Pipeline *LangChain • CrewAI • OpenAI APIs*

- Developed a scalable multi-agent architecture for extracting and synthesizing knowledge from entire YouTube channels.
- Automated transcript ingestion, semantic retrieval, topic modeling, and long-form content generation.
- Reduced manual research effort by approximately 90%.

Education

VIT Bhopal University

Sep 2022 – Jul 2026 Bachelor of Technology in Computer Science Engineering

CGPA: 7.71

Achievement: Meta HackerCup 2025 Finalist (AI Track)

Certifications: MongoDB Administrator — FacePrep — Generative AI with LangChain — Udemy — Computer Vision Deep Learning — Udemy

Technical Skills

Programming: Python, C++, SQL

Machine Learning: Scikit-learn, TensorFlow, Keras, supervised learning, model evaluation

Generative AI: Retrieval-Augmented Generation (RAG), LangChain, vector databases (FAISS, Neo4j), prompt engineering

Data & Tools: Pandas, NumPy, Git, MongoDB, Streamlit